Example Delayed Exam

Analytical Skills for Business

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Exam Information

This document contains an alternative delayed exam for the Analytical Skills for Business course at Hochschule Fresenius.
Total Points: 90
Duration: 90 minutes (1 point per minute)
Questions: 7 sections

Collaborative Analytics Workflow and Reproducibility

Points: 14

Question

a) (6 points) In an analytics team, two analysts are working on the same forecasting project. Describe a simple collaboration workflow with Git branches and pull requests.

b) (4 points) Explain briefly how commit messages and repository history support reproducibility in business analytics projects.

c) (4 points) Give two concrete risks for analytics projects when version control is not used.

Data Types and Suitable Visualizations

Points: 16

Question

a) (8 points) Distinguish between categorical, numerical, and time series data. For each type, name one suitable visualization and explain why it fits.

b) (8 points) A company wants to communicate weekly sales performance to PR managers. Propose two visualizations and explain which business question each visualization answers.

Data Quality Management Process

Points: 14

Question

a) (6 points) Describe the five-step data cleaning workflow (profiling, validation, cleaning, verification, documentation) and explain the goal of each step.

b) (8 points) Choose two of the following data issues and explain one detection method plus one treatment strategy for each: missing values, inconsistent formats, duplicate records, outliers.

Association and Model Interpretation

Points: 14

Question

a) (7 points) Explain what correlation measures.

b) (7 points) Give one business example where correlation is useful and one business example where linear regression is useful.

Regression Quality and Assumptions

Points: 12

Question

a) (6 points) Explain the meaning of R-squared in simple terms.

b) (6 points) Name an assumption of linear regression and explain a practical consequence if theis assumption is violated.

No-Code/Low-Code in Analytics Strategy

Points: 10

Question

a) (5 points) Select two tools from the course (for example n8n, Tableau, Power BI, KNIME, QGIS, Streamlit on Snowflake). For each tool, describe one realistic business use case.

b) (5 points) Explain one situation where no-code/low-code is preferable to custom programming, and one situation where custom programming is preferable.

Practical R Programming and Data Wrangling

Points: 10

Part a: Tibble and Summary

Points: 4

Question

Write R code to create a tibble named `kpi_data` with two columns:

- `month` = Jan, Feb, Mar, Apr
- `revenue` = 42, 50, 47, 55

Then calculate and print:

- mean revenue
- standard deviation of revenue

Part b: Percentiles

Points: 3

Question

Write R code to compute the 25th, 50th, and 75th percentiles of the `revenue` column in `kpi_data`.

Part c: Simple Visualization

Points: 3

Question

Write R code with `ggplot2` to create a line chart of revenue by month.

References

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